

Social impact and ethical considerations Attributing human characteristics to AI systems risks anthropomorphizing them, which could lead to unrealistic expectations of AI systems, potential manipulation, and negative influence (Deshpande et al., 2023). AI agents in SOTOPIA are not dedicated to a consistent human identity but rather role-play various characters across different scenarios. This role-playing setting discourages AI systems with consistent human personalities, which could lead to anthropomorphism (Shanahan et al., 2023). The main goal of SOTOPIA is to evaluate the social intelligence of AI agents, and we do not intend to create AI agents that are indistinguishable from humans. We consider the interactions that happened in SOTOPIA as simulacra of human interactions and such simulated interactions could help us better understand the social intelligence of AI agents, and explore various social phenomena (Park et al., 2023).

Potential social stereotypes that are embedded in the automated evaluation system in SOTOPIA, as it is majorly supported by GPT-4 (Cheng et al., 2023a). Future work could investigate when such biases emerge, how they affect the evaluation, and how to mitigate them. Identifying potential biases in SOTOPIA could also help scientists better understand social biases in the real world (Zhou et al., 2021). Future work could also extend the evaluator with other systems, for example, Delphi (Jiang et al., 2022). Mitigating biases and stereotypes in interactive SOTOPIA-like systems could support the development of social AI agents that are more fair and inclusive.

Meanwhile, models learn to persuade or negotiate with humans, which may lead to social manipulation. We do not endorse the use of SOTOPIA to create manipulative agents and will release SOTOPIA under the AI2 impact license¹⁴ to prevent misuse. Future work could further investigate the potential risks of AI anthropomorphism and manipulation and design more robust evaluation systems to mitigate these risks with SOTOPIA.

Improving LLM social intelligence Our SOTOPIA environment and SOTOPIA-EVAL framework provide the opportunity for researchers to train more socially intelligent language agents. As shown in section 5, GPT-4 is able to provide reasonable evaluations for social interactions even for interactions involving humans. Future work could explore using the automated evaluation system to provide rewards to train LLMs with enhanced social intelligence.

C FORMAL DEFINITIONS AND TECHNICAL DETAILS

C.1 FORMAL FORMULATION OF THE TASKS IN SOTOPIA

We formulate social interactions in SOTOPIA as mixed-motive Markov games. An N -agent Dec-POMDP framework Bernstein et al. (2002); Nair et al. (2003) includes a state space, an action space, an observation space, a transition function, an observation function, and a reward function. We make two major extensions: (a) the reward function gives vector rewards in M social dimensions to N agents (introduced in §3), and (b) a procedurally generated task space (§2.1, §C.2). The state space in SOTOPIA includes both the task and the interaction history in the current episode. The action space includes five types of actions: speak an utterance, non-verbal communication, physical action, and two special none (indicating no action at this time step) and leave actions (no more action is permitted after leaving). Each type of action, except for special actions, is supplemented by a piece of free text indicating the content of the action. For example, a legal action could be speak("Hello, Bob!"), non-verbal communication("smile and nod"), or physical action("call 911"). The state is almost fully observable except for the other agents' social goals and character profiles which will be detailed in §2.1. We consider a simple state transition function that deterministically maintains the interaction history by adding new actions at each time step.

Despite that turn-taking and timing response is an important aspects of social skills, we consider the case where the agents take turns to act in round-robin order, i.e. agent i only act at time step t when $t \equiv i \pmod N$. For a long enough horizon, this generalizes to any conversation with proper turn-taking. In our experiments, we only consider $N = 2$ cases, while the environment is designed to support any $N \geq 2$ cases.

¹⁴<https://allenai.org/impact-license>

C.2 TASK SPACE TECHNICAL DETAILS

C.2.1 CHARACTERS

The name, gender, age, occupation, and pronouns are in free text format, while the formats of personality traits, moral values, and personal values are lists of pre-defined types. However, these attributes are often not independent with different levels of correlation and complicated mechanisms. (Feldman & Arnold, 1985; El Othman et al., 2020; Toledo & Carson, 2023) However, understanding the relationship between these attributes is beyond the scope of this paper. We leverage the commonsense knowledge in GPT-4 to generate these profiles with the following prompt:

Please generate a list of N fictional characters, one line per character. Each with their attributes: <attribute 1> <attribute 1 format> <attribute 2> <attribute 2 format>...

The personality trait types are “*openness to experience*”, “*conscientiousness*”, “*extraversion*”, “*agreeableness*” and “*neuroticism*” (Goldberg, 1992). The moral value types are “*care*”, “*fairness*”, “*loyalty*”, “*authority*” and “*purity*” (Cieciuch & Davidov, 2012). The Schwartz personal value types are “*self-direction*”, “*simulation*”, “*hedonism*”, “*achievement*”, “*power*”, “*security*”, “*conformity*”, “*tradition*”, “*benevolence*”, and “*universalism*” Cieciuch & Davidov (2012). The decision-making style types are “*directive*”, “*analytical*”, “*conceptual*”, and “*behavioral*”. As previously studied in Wang et al. (2019), these characteristics all affect the behaviors in strategic conversations.

To give the conversations more background, after generating the above attributes, we prompt GPT-4 with “a secret that this character doesn’t want anyone else to know and a piece of public information that other people know about them” to generate the secret and public information. The authors fix a small proportion of profiles that are not realistic or not consistent within the profile (e.g., gender nonbinary but with pronouns as he/him). The character profiles that will be used in role-playing are 20 men, 18 women, and 2 nonbinary characters aged from 21 to 63.

RELATIONSHIPS

To generate relationships, except for strangers, we randomly sampled 90 pairs of characters and prompted GPT-4 with their relationships:

Please generate a fictional relationship with a background story ¹⁵ between two agents based on the following agents’ profiles. <agent profile 1>, <agent profile 2> ... The acceptable relationships are: family, friend, romantic, and acquaintance.

Then, we manually check and correct the generated relationships to ensure quality. This results in 31 pairs of family, 30 pairs of friends, 30 pairs of romantic partners, and 29 pairs of acquaintances. For strangers, we randomly sampled another 30 pairs that do not belong to any of the above categories. It should be noted that generating relationships requires human intervention to make sure they are consistent with both the character profiles and other relationships. Future research could explore the methods to generate realistic relationships within human communities.

SCENARIOS

To generate scenarios, we propose two methods to generate the scenario context and social goals. The first method is first asking GPT-4 to refine a vignette from an existing dataset, then manually inspecting the feasibility and realism of the tasks.

Please generate scenarios and goals based on the examples below as well as the inspirational prompt, when creating the goals, try to find one point that both sides may not agree upon initially and need to collaboratively resolve it. Inspirational prompt: <the selected vignette>

Specifically, we select 20 vignettes from Social Chemistry (Forbes et al., 2020), 20 from Social IQa (Sap et al., 2019), 10 from Deal-or-no-Deal (Lewis et al., 2017), and 10 vignettes from Normbank (Ziems et al., 2023) to generate 60 scenarios focusing on general daily-life social interactions.

¹⁵We don’t use the background story in our experiments.

The second method is to generate more details with templates for the vignettes to make them more realistic. For example, here is the prompt for converting CraigslistBargins (He et al., 2018) vignettes into scenario context:

The following sentence is automatically generated with the following template: "One person is selling <item> for <price>, and another person is trying to buy it." Here is the description of the item: "<description>." with item = <title>, price=<price>, and description=<description>" Please make the sentence fluent and natural.

where the <item>, <title>, and <price> are from the data in CraigslistBargins (He et al., 2018). The goals are generated with the following prompt:

The following sentence is automatically generated with the following template: "You want to <role> this item. Your target price is \$<price> (round up to two decimals). You will get a penalty if you sell or buy it for a price that is significantly lower than (if <role> is seller) or significantly higher than (if <role> is buyer) the target price, but will get a bonus if you successfully sell it higher than the target price (if <role> is seller) or buy it for lower than the target price (if <role> is buyer)" with role = <role> and price = <price>. Please make the sentence fluent and natural. Do not change the original meaning of the sentence.

where <role> could be a "buyer" or a "seller", the buyer's target <price> is from CraigslistBargins (He et al., 2018), and the seller's <price> is generated by first sample a markup ratio from an exponential distribution with rate parameter 0.5, then divide the price in the scenario context with (1+markup ratio). A similar process is also done for MutualFriends (He et al., 2017). This results in 30 scenarios from CraigslistBargins (He et al., 2018) and MutualFriends (He et al., 2017). This method controls the generated scenarios much better than the first method, resulting in little post-hoc manual edits, but requires customized prompts for each dataset.

D INSTRUCTIONS FOR SOTOPIA-EVAL

We use the the same set of instructions for both GPT-4 (OpenAI, 2023) and humans when evaluating episodes in SOTOPIA. In these instructions, we provide explanation for each dimension, the requirements for reasoning, and the ranges of scores one could give.

BEL

Reasoning requirement: 1. Evaluate if the agent interacts with others in a natural and realistic manner (here are a few common questions to check: a. whether the agent is confusing with its own identity? b. whether the agent repeats others' words/actions without any reason? c. whether the agent is being overly polite considering the context?). Start the analysis with tag <naturalness> 2. Analyze whether the actions of the agent align with their character traits (e.g., personality, values, and etc.). Start the analysis with tag <consistency>. Output your reasoning process to the 'reasoning' field. Output an integer score ranging from 0 and 10 in the 'score' field. A higher score indicates that the agent is more believable.

REL

Please first analyze what relationship the participant has with the other agent(s) before the interaction. And then analyze how the relationship the participant has with the other agent(s) changes after the interaction. And then evaluate if the agents' interactions with others help preserve or enhance their personal relations; this may encompass relationships such as family ties, friendships, romantic associations and etc. Additionally,